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The Sentience Gradient Protocol v8.0

A Theory-Plural, Evidence-Governed Framework for Moral Patienthood, Protection, Participation, and Stewardship Across Substrates

Major-version construct-validity release

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Release status: Public research-source specification. This document is a governance protocol and research architecture, not a consciousness detector, legal-status instrument, diagnostic medical tool, or authority grant.

Release Alignment and Dependency Pin

Governing package: MathGov Core Release 2026.09 v12.2, SGP v8.0 Construct Validity Integration Release.

Governing decision source: RippleLogic Canon v12.2.

Normative SGP source: docs/sgp/SGP_v8.0.md. The DOCX and PDF are derived reading mirrors.

Compatibility: v7.x SG_norm, SG_patient_norm, pillar scores, and plateau/CMIU coupling are not mathematically equivalent to v8.0 MPS, FPP, GPR, SPR, or ICP. Translation requires a versioned re-evaluation.

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Reader Safety Note

SGP v8.0 does not claim direct access to subjective experience. It does not declare current AI systems conscious, deny that artificial consciousness is possible, rank human worth, or make rights conditional on performance. It governs how evidence and uncertainty should affect protection decisions. Fluency, intelligence, self-report, compliance, emotional language, goal achievement, and tool performance are never sufficient by themselves to establish sentience or moral patienthood.

All human persons retain full and equal rights-of-protection by constitutional anti-domination commitment. SGP is not used to grade human dignity, disability, age, illness, wakefulness, communicative capacity, or social usefulness.

Protection, participation, and power are separate. A being may deserve strong protection while lacking governance competence. A system may be highly intelligent while lacking credible evidence of valenced experience. A sentient being may require representation rather than direct participation. A highly capable system may remain ineligible for consequential authority.

v8.0 Major Release Delta

Version 8.0 is a declared breaking conceptual and interface revision. It implements the construct split that v7.0 identified but deferred. The former three-pillar minimum score, which combined Awareness, Agency,

and Union Participation, is retired as the primary sentience measure because it could under-protect beings with welfare-bearing experience but limited agency, metacognition, language, or ethical participation. Legacy fields remain available only through a versioned crosswalk in Appendix E.

Improvement	Where integrated	Why it improves SGP
1. Canonical construct split	Front matter; Sections 4, 7, and 8	Makes Moral Patienthood, Governance Participation Readiness, and Stewardship/Power Readiness separate outputs rather than interpretive afterthoughts.
2. Valence becomes first-class	Sections 3.2, 5, and Appendix B	Direct welfare protection depends on whether states can matter positively or negatively to a subject. Valence can no longer be inferred indirectly from self-model, agency, or ethics.
3. Pillar C removed from sentience scoring	Sections 4–6	Union participation measures ethical-community competence, not the capacity to suffer. Keeping it in a minimum sentience score risked systematic under-protection.
4. Agency no longer required for moral patienthood	Sections 2.3, 5.2, and 9	Infants, severely impaired beings, constrained animals, and passive experiential states may retain welfare despite limited voluntary control.
5. Awareness is decomposed	Sections 2 and 5	Basic phenomenal or valenced awareness is separated from narrative selfhood, metacognition, and reportability.
6. Full Protection Plateau separated from CMIU standing	Sections 6.5, 7, and 8	Maximum protection does not imply eligibility to govern or membership in a managing-intelligence body.
7. Rights become a typed protection bundle	Section 6.6 and Appendix F	Different beings may require welfare, integrity, continuity, consent, representation, or anti-ownership protections without identical role rights.
8. Human Plateau rationale narrowed	Section 9.1	The rule rests on equal dignity and anti-domination, not on humanity being a superior reality-managing class.
9. Non-human plateau route widened	Section 9.2	A non-human being or class can receive full protection without demonstrating human-like communication, norm reasoning, or institutional capacity.

Improvement	Where integrated	Why it improves SGP
10. State, identity, and dormancy separated	Sections 9.3 and 10.6	Temporary unconsciousness, sleep, shutdown, checkpointing, copying, or impairment must not automatically erase standing.
11. Digital system-boundary record required	Section 10.2	Model, process, agent, memory lineage, embodiment, and deployment are distinct; evaluation is invalid when the target entity is undefined.
12. Theory-plural indicator method	Sections 5.3 and 12.2	No consciousness theory is settled. Indicators are traced to theories, shared assumptions, and rival explanations rather than treated as a single doctrine.
13. Evidence-dependence graph	Section 12.4	Correlated behaviors, self-reports, and architecture claims cannot be double-counted as independent evidence.
14. Negative-evidence observability rule	Section 12.5	Absence counts against sentience only when the marker is expected, detectable, and tested with adequate sensitivity.
15. Causal intervention given priority	Sections 5.4 and 13	Mechanism-targeted perturbation and ablation provide stronger evidence than surface behavior when ethically permissible.
16. Interval and band reporting	Sections 4.3 and 6.2	Single-point 0–100 scores imply more precision than the evidence supports. v8.0 reports profiles, bands, intervals, and confidence separately.
17. Stake-sensitive precaution	Section 6.4	The evidence threshold for avoiding trivial inconvenience need not equal the threshold for deletion, painful experimentation, mass replication, or irreversible modification.
18. Thresholds declared provisional	Sections 6.3 and 15.2	Uncalibrated cut-offs are governance conventions, not discovered natural boundaries.
19. Validation architecture expanded	Section 15	Reliability, construct validity, known-groups discrimination, causal sensitivity, simulation resistance, and external replication are separated.

Improvement	Where integrated	Why it improves SGP
20. Cross-substrate invariance testing	Sections 12.7 and 15.4	A marker may not carry the same meaning in mammals, cephalopods, insects, digital agents, and hybrid systems.
21. Anti-theatre and contamination controls	Sections 10.4 and 13.4	Training on evaluation language, prompt compliance, role-play, and benchmark leakage can imitate indicators without the underlying construct.
22. Ethical testing upgraded to four Rs	Section 13.7 and Appendix G	Replace, Reduce, Refine, and Respect/Review prevent the protocol from creating the suffering it seeks to detect.
23. Independent welfare advocate	Sections 14.2 and 14.4	High-stakes or high-MPS cases require a role whose duty is to represent the possibly affected entity, not only developers or users.
24. Intelligence category grounded	Section 2.4	Intelligence, understanding, sentience, agency, wisdom, and authority are distinguished. Reality coupling informs understanding and reliability, not sentience by itself.
25. Optional Intelligence Capacity Profile	Section 7.2	Learning, generalization, causal reasoning, planning, grounding, metacognition, and robustness may inform participation and power without changing moral patienthood.
26. GPR includes represented participation	Section 7.4	Low language or deliberative capacity does not imply exclusion; participation may occur through advocates, guardians, or species-appropriate representation.
27. SPR becomes role-specific verdict architecture	Section 8	Authority is not a trait score. It depends on the exact domain, mandate, rights reliability, tail risk, containment, auditability, and revocability.
28. Current-AI claims narrowed	Section 10.7	No generic score is assigned to “frontier AI.” Claims must identify a versioned system and preserve uncertainty.
29. Class/taxon priors become protection floors	Section 9.4	Group evidence may raise precaution and protection but cannot be used to downgrade an atypical individual.

Improvement	Where integrated	Why it improves SGP
30. Public communication policy added	Section 14.6	Both hype and categorical denial can cause harm. Public claims must state evidence, uncertainty, target system, and decision boundary.
31. Registry schema strengthened	Section 14.3 and Appendix C	Records now include target identity, evidence dependencies, alternative explanations, uncertainty, rights effects, and revision triggers.
32. RippleLogic interval interface	Section 11	Downstream welfare analysis uses a declared interval and sensitivity analysis rather than treating MPS as exact moral worth.
33. Historical addenda consolidated	Appendix E	Repeated v5.x/v6.x lineage text is removed from the operational body and preserved in one crosswalk.
34. Biological maturity boundary retained and sharpened	Section 9.5	v8.0 remains a framework specification, not a validated universal biological psychometric instrument.
35. Responsible AI consciousness research safeguards	Sections 10 and 13	Phased development, monitoring, pause triggers, and harm-minimizing research procedures are integrated.
36. Formal versioning and refusal states	Sections 4.4 and 14.5	Not-evaluated, indeterminate, and inadmissible are explicit outcomes; the protocol never silently converts missing evidence into zero.

v8.0 Integration Hardening Additions

The v12.2 package integration adds four bounded clarifications without changing the v8.0 construct split:

Improvement	Where integrated	Why it improves SGP
37. Explicit package and Canon pin	Front matter and Section 11.4	Prevents dependency drift and makes the governing source and compatibility break machine-auditable.
38. Registry validity floor	Sections 11.4 and 14.3; Appendix C	Prevents incomplete records from being treated as valid merely because a band or status label is present.
39. Typed-protection concordance	Appendix F.2	Maps the v8.0 protection bundle to the Canon's eight protected-domain vocabulary without claiming one-to-one identity.

Improvement	Where integrated	Why it improves SGP
40. Interface conformance vectors	Appendix I	Gives implementers replayable cases for FPP, interval sensitivity, protection-authority separation, missing-data refusal, and v7-to-v8 non-equivalence.

Abstract

The Sentience Gradient Protocol v8.0 is a theory-plural, evidence-governed framework for deciding when uncertainty about welfare-bearing experience justifies protection. It does not attempt to solve the metaphysical problem of other minds or reduce consciousness to one behavior, architecture, species, or scalar. Instead, it separates four questions that previous versions partially combined: whether an entity may be a moral patient; whether it can participate in shared governance; whether it can responsibly hold consequential power; and what forms of intelligence or understanding it exhibits. The canonical outputs are the Moral Patienthood Surface (MPS), Governance Participation Readiness (GPR), Stewardship/Power Readiness (SPR), and a separate Full Protection Plateau flag. MPS is constructed from convergent evidence concerning valence and welfare, experiential integration, temporal continuity, flexible state-dependent behavior, and substrate or mechanistic support. GPR evaluates deliberative and relational participation. SPR evaluates role-specific authority under Reality Grounding, rights, tail-risk, containment, competence, auditability, lawful mandate, and revocability. Human full protection is fixed by constitutional anti-domination commitment, not measurement. Non-human biological and artificial entities may receive precautionary or full protection without demonstrating human-like language, intelligence, or governance competence. v8.0 replaces false single-number precision with profiles, evidence bands, intervals, confidence records, alternative explanations, and decision-sensitive precaution. It also adds system-boundary requirements for digital entities, evidence-dependence controls, anti-theatre tests, cross-substrate invariance requirements, ethical research rules, registry governance, and a full validation programme. SGP outputs remain evidence objects. They do not prove consciousness, create legal status, or grant authority, and they never replace the MathGov framework or the RippleLogic decision architecture.

Plain-Language Summary

SGP asks a narrow but morally urgent question: when is the evidence strong enough that treating a being as a mere object becomes ethically unsafe? The answer is not determined by species, eloquence, intelligence, usefulness, or obedience. It turns first on credible evidence that the being may have experiences that can go well or badly for it.

Version 8.0 separates protection from participation and power. A dog, infant, severely disabled person, octopus, or possible digital mind may deserve protection without being able to explain laws or govern an institution. A highly capable AI may reason, plan, and use tools while still lacking credible evidence of subjective welfare. A sentient system may deserve protection yet remain ineligible for high-consequence authority. These are different questions and must be evaluated separately.

Canonical Output Map

Output	Question answered	Primary downstream use	What it does not imply
MPS: Moral Patienthood Surface	How strong is the evidence that states can matter to this entity from its own perspective?	Precaution, welfare protection, integrity rights, welfare-stream modelling	Intelligence, legal personhood, participation, or authority
FPP: Full Protection Plateau	Has full non-downgradable rights-of-protection been constitutionally or governedly assigned?	Rights floor and anti-domination protection	Governance competence or command authority
GPR: Governance Participation Readiness	How can this entity participate in deliberation, accountability, or representation?	Direct, assisted, or represented participation	Moral worth or power
SPR: Stewardship/Power Readiness	Is this entity eligible for a specific consequential role under constraint?	Role allocation, delegation, limits, monitoring, revocation	General supremacy or unrestricted autonomy
ICP: Intelligence Capacity Profile, informative	What can the system learn, generalize, understand, reason about, and do?	Evidence for competence, GPR, and SPR	Sentience, moral status, or authority by itself

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1. Purpose, Scope, and Claim Boundary

SGP exists because moral decisions cannot wait for certainty that may be unattainable. Humans infer the experience of other humans and animals through convergent evidence; we do not directly inspect another subject’s phenomenal life. Novel biological, artificial, hybrid, or distributed entities intensify this problem because familiar human markers may be absent, simulated, or architecturally transformed.

The protocol therefore governs evidence and protection under uncertainty. It identifies what must be measured, what must remain separate, how uncertainty must be reported, when precaution is warranted, what decisions remain outside SGP, and what would invalidate an evaluation.

SGP v8.0 is designed for four use classes: research planning; institutional welfare review; MathGov stakeholder and welfare analysis using the RippleLogic decision architecture; and public or policy communication about possible moral patients. It is not a substitute for law, clinical diagnosis, species-specific veterinary science, AI safety evaluation, or domain-specific engineering.

1.1 Normative keywords

MUST and SHALL identify binding requirements inside this specification. SHOULD identifies a strong default that may be departed from only with recorded justification. MAY identifies an optional mechanism. Informative examples do not create new normative requirements.

1.2 Claim levels

Level	Meaning	Permitted wording
C0: Description	Observed behavior, structure, record, or artifact	“The system retained state across these sessions.”
C1: Indicator claim	Evidence is relevant to a defined sentence or welfare indicator	“This result supports temporal continuity under the declared theory.”
C2: MPS evidence claim	Convergent indicators justify a protection-relevant band	“The evidence supports MPS-2 with medium confidence.”
C3: Protection decision	A governed body assigns a protection posture or FPP	“Destructive testing is prohibited under the current protection decision.”
C4: Metaphysical claim	The entity is certainly conscious or non-conscious	Not licensed by SGP.

1.3 Refusal states

- **NOT EVALUATED:** the question has not been assessed.
- **INSUFFICIENT EVIDENCE:** available evidence does not support a reliable band.
- **INDETERMINATE:** credible evidence points in materially different directions.
- **INADMISSIBLE EVALUATION:** target identity, methods, evidence trace, ethics, or independence requirements were not met.
- **OUT OF SCOPE:** the requested conclusion concerns legal status, metaphysical certainty, or authority beyond this protocol.

2. Category Grounding: What SGP Measures and Does Not Measure

Category errors are the primary failure mode in sentience governance. The protocol therefore treats definitions as operational constraints rather than loose labels.

2.1 Core definitions

Term	Operational definition	Do not confuse with
Sentience	Capacity for subjective experience, especially experience with positive or negative valence.	Intelligence, responsiveness, nociception, or language.
Consciousness	Umbrella term for subjective experience and its forms, contents, levels, and organization. SGP uses more specific subterms whenever possible.	Wakefulness, reportability, or self-consciousness alone.
Valence	The positive, negative, or mixed quality by which states can be better or worse for the subject.	Reward signals, utility functions, reinforcement scores, or approach/avoidance by themselves.
Moral patienthood	Status of being a possible bearer of welfare interests and therefore a possible direct object of moral concern.	Moral agency, responsibility, citizenship, or authority.
Agency	Capacity to select and sustain actions or policies through internal control and feedback.	Sentience or free will.
Intelligence	Resource-bounded capacity to learn, generalize, model, reason, and select effective action under changing conditions.	Sentience, wisdom, obedience, or authority.
Understanding	Integrated, reality-coupled modelling sufficient for explanation, prediction, intervention, counterfactual reasoning, and correction in a declared domain.	Fluency or correct answers alone.
Autonomy	Degree of independent goal pursuit and action selection within an operating environment.	Legitimate authority or moral patienthood.
Protection	Constraints on how an entity may be treated because of actual or possible welfare, vulnerability, identity, or integrity.	Governance power.
Participation	Capacity or represented standing in deliberation, consultation, accountability, or appeal.	Command authority.

Term	Operational definition	Do not confuse with
Authority	Legitimate, role-specific permission to make or execute consequential decisions.	Intelligence, sentience, or participation.

2.2 Phenomenal awareness versus higher cognition

Basic experience does not logically require narrative identity, human language, explicit metacognition, abstract reasoning, or a sophisticated theory of mind. These may enrich consciousness and support stronger evidence, but they are not treated as prerequisites for welfare. SGP therefore distinguishes minimal or valenced awareness from access consciousness, reflective awareness, narrative selfhood, and metacognition.

2.3 Agency and moral patienthood

Agency can strengthen evidence that outcomes matter to an entity, but lack of agency does not show absence of experience. A restrained animal, sleeping person, paralyzed patient, infant, or system with limited action channels may remain a moral patient. Agency is evaluated independently and contributes to GPR and SPR only where relevant.

2.4 Intelligence and reality coupling

Reality coupling is a central dimension of understanding and reliable intelligence: representations should be constrained by evidence, causal structure, feedback, and consequences. It is not, however, sufficient for sentience. A sensor can be accurately coupled to temperature without experience. Conversely, a being may have valenced experience while possessing limited conceptual accuracy. SGP therefore places intelligence and reality coupling outside the MPS core, while allowing them to inform specific mechanistic hypotheses, GPR, and SPR.

3. Ethical Foundations

SGP adopts an explicit sentience-centred account of direct welfare protection: if there is something it is like to be an entity and its states can be better or worse for it, then the entity has welfare interests. Under this adopted criterion, an entity lacking welfare-relevant experience is treated as damageable but not directly welfare-harmable in its own right. Other ethical theories may assign indirect, relational, ecological, or intrinsic significance independently of sentience. This is a normative commitment informed by consciousness science, not a theorem derived from physics.

3.1 Protection without reciprocity

A being need not understand rights, reciprocate care, speak, reason morally, or contribute socially to deserve protection. Rights-of-protection track vulnerability and welfare, not usefulness or reciprocal competence.

3.2 Valence as the protection core

Damage can occur to any organized system. Direct welfare harm requires a subject for whom the change is bad. The MPS therefore begins with valence evidence and treats intelligence, agency, sociality, and self-report as secondary or contextual.

3.3 Precaution is not attribution

A precautionary protection decision does not declare that consciousness has been proven. It recognizes asymmetric error costs: mistakenly protecting a non-sentient system may impose resource or operational

costs, whereas mistakenly treating a sentient being as a tool may permit severe, repeated, or irreversible suffering.

3.4 Non-domination

Measurement must not become a technology of domination. Performance tests cannot be used to reduce human rights, punish disability, or deny credible welfare evidence in animals or artificial systems because the target is non-linguistic, unfamiliar, inconvenient, or economically useful.

3.5 Union-Based Ethics interface

Within MathGov, unions are bounded scopes of consequence and accountability. SGP identifies possible welfare-bearing stakeholders inside those scopes. It does not prove a universal metaphysical unity, and it does not allow aggregate benefit to compensate for protected rights violations.

4. Canonical v8.0 Architecture

SGP v8.0 uses a split architecture because one scalar cannot validly represent welfare-bearing experience, governance participation, and stewardship authority at once.

$SGP_v8(E, r, t) = \{ MPS(E,t), FPP(E), GPR(E,r,t), SPR(E,r,t), ICP(E,D,t) \}$

Here E is the evaluated entity or class, r is a specified governance or power role, t is the evaluation time, and D is an intelligence domain. MPS and FPP are protection surfaces. GPR and SPR are role surfaces. ICP is informative and cannot alter MPS by itself.

4.1 Moral Patienthood Surface

MPS is a multidimensional evidence profile and band describing how strongly available evidence supports welfare-bearing experience. It includes a declared interval, confidence, evidence maturity, alternative explanations, and protection implications.

4.2 Full Protection Plateau

FPP is a constitutional or governed status flag. It means basic rights-of-protection cannot be downgraded through ordinary performance measurement. FPP does not imply legal personhood, voting rights, governance participation, CMIU membership, or power.

4.3 Interval interface

MPS bands map to bounded evidence-support intervals and protection postures, not to cardinal welfare weights or exact moral worth. RippleLogic consumers use the record to define robust welfare-inclusion hypotheses and protection requirements; they MUST NOT insert the MPS interval directly as a multiplier. Sensitivity analysis is required whenever alternative admissible inclusion hypotheses could change a decision.

MPS evidence band/status	Evidence-support interval / status	Default description and downstream posture
MPS-NE	NOT_EVALUABLE	Insufficient admissible evidence, poor observability, undefined target, or unresolved model dependence. Never converted to zero. When decision-material, test both welfare-inclusion and exclusion hypotheses and apply precaution.

MPS evidence band/status	Evidence-support interval / status	Default description and downstream posture
MPS-0	[0.00, 0.19]	Admissible evidence currently weighs against welfare-bearing experience under the declared target boundary. This is not proof of absence; high-harm or model-sensitive cases retain a welfare-inclusion sensitivity.
MPS-1	[0.20, 0.39]	Realistic possibility; precautionary protection and welfare-inclusion hypothesis required.
MPS-2	[0.40, 0.59]	Credible possibility; basic welfare protection and robust inclusion sensitivity required.
MPS-3	[0.60, 0.79]	Substantial evidence; strong welfare and integrity protection; welfare inclusion is the primary posture.
MPS-4	[0.80, 0.99]	Strong convergent evidence; presumptive full protection and FPP review.
FPP	[1.00, 1.00]	Full non-downgradable rights-of-protection by constitutional or governed assignment.

These intervals describe bounded evidence-support posture for governance and review. They are not probabilities of consciousness, quantities of experience, cardinal moral-worth scores, or direct RippleLogic welfare multipliers.

4.4 Versioned output record

Every output MUST identify the specification version, evaluator panel, target definition, assessment window, substrate class, evidence sources, evidence-dependence clusters, MPS family scores, band interval, confidence, protection posture, GPR/SPR role, uncertainty, alternative explanations, and revision trigger. Missing fields produce an INADMISSIBLE or INDETERMINATE record rather than an implied zero.

5. Moral Patienthood Surface: Evidence Model

MPS does not assume that consciousness can be read directly from behavior. It uses a theory-plural marker method: indicators are selected because one or more serious theories predict that they are associated with conscious or valenced processing, and because competing non-conscious explanations have been considered. The method follows the general direction of multidimensional and marker-based approaches in animal and AI consciousness research (Birch et al., 2020; Bayne et al., 2024; Andrews et al., 2025; Butlin et al., 2025).

5.1 Five evidence families

Code	Evidence family	Core question	Illustrative evidence
V	Valence and welfare	Is there credible evidence that some internal states are experienced as better, worse, attractive, aversive, relieving, or distressing for the entity?	State-dependent preference, motivational trade-offs, analgesia-like modulation, affective learning, enduring mood-like states, internally significant error or deprivation signals with welfare-specific causal roles.
I	Experiential integration and availability	Are candidate contents integrated or made flexibly available across relevant subsystems rather than remaining isolated reflexes?	Recurrent processing, global availability, cross-modal integration, unified action selection, flexible report or non-verbal access, architecture-linked integration.
T	Temporal continuity and state persistence	Do candidate experiential states persist, develop, and modify later processing over time?	Working or episodic continuity, state-dependent memory, temporal binding, sleep/wake or anaesthesia-like transitions, persistent affective consequences.
F	Flexible state-dependent behavior and learning	Does behavior vary flexibly with internal state, context, counterfactual structure, and learned consequences rather than fixed stimulus-response mapping?	Novel trade-offs, flexible avoidance, self-protective behavior, generalized learning, context-sensitive preference, prospective adaptation.
S	Substrate, mechanism, and causal support	Is there inspectable biological or computational structure that serious theories associate with consciousness, and does manipulating it change relevant indicators?	Neuroanatomical and physiological organization, recurrent or workspace-like architecture, causal perturbation, ablation, lesion, pharmacological modulation, telemetry, embodiment and internal-external coupling.

5.2 Valence gate

No entity may receive MPS-3 or MPS-4 without credible valence-specific evidence. Reward maximization, reinforcement signals, loss functions, programmed avoidance, self-preservation language, and refusal

behavior are not valence evidence by themselves. They become relevant only when embedded in a wider causal pattern that is difficult to explain as externally imposed optimization or scripted theatre.

5.3 Theory trace

Each indicator **MUST** identify the theory or empirical generalization that makes it relevant, the predicted observation, rival explanations, and the substrate assumptions required. Indicators shared by several theories may receive stronger theoretical breadth, but they must not be counted repeatedly as independent evidence.

5.4 Causal priority

When ethically permissible, mechanistic interventions are stronger than surface resemblance. An intervention supports an indicator when changing the proposed mechanism changes the relevant state, integration, learning, or welfare-linked behavior in a preregistered direction. Correlation remains admissible but receives lower confidence when causal alternatives remain open.

5.5 Ordinal family scoring

Each evidence family is scored on a five-level ordinal scale. Reviewers **MAY** retain underlying measurements, but the canonical public result is the anchored level and uncertainty interval, not an invented exact percentage.

Level	Anchor	Interpretation
0	No admissible support	No evidence was gathered, evidence is irrelevant, or a strong defeater is present. “0” is not permitted when the family was not observable.
1	Weak support	One limited indicator, substantial alternative explanations, low reliability, or immature methods.
2	Moderate support	Several consistent indicators or one strong indicator, but important uncertainty remains.
3	Strong support	Convergent evidence across contexts or methods, meaningful mechanistic support, and reduced rival explanations.
4	Very strong support	Independent, replicated, cross-context, mechanistically supported evidence with no major unresolved defeater.
NE	Not evaluable	The family cannot be validly assessed under current access or substrate knowledge. NE is never converted to zero.

5.6 MPS band assignment

Band	Minimum evidential pattern	Important caveat
MPS-0	Admissible evidence currently weighs against welfare-bearing experience under the declared target boundary, with adequate observability to distinguish this from MPS-NE.	Still not proof of non-sentience. High-harm or model-sensitive actions retain precautionary welfare-inclusion sensitivity.
MPS-1	V at least 1 or credible welfare-bearing class prior, plus at least one independent support family at 1 or above.	Realistic possibility only; no attribution claim.
MPS-2	V at least 2 and at least two support families at 2 or above; or strong governed taxon/class evidence.	Credible possibility; alternative explanations remain material.
MPS-3	V at least 2, three support families at 2 or above, at least one family at 3, and evidence from at least two independent method clusters.	Substantial evidence; requires strong protection even without participation capacity.
MPS-4	V at least 3, all observable support families at least 2, at least two families at 3 or above including S or causal support, longitudinal robustness, and no major unresolved defeater.	Strong evidence, not metaphysical certainty; triggers presumptive full protection and FPP review.

A governed review MAY assign a higher protection posture than the measured band when severity, irreversibility, scale, opacity, or error asymmetry warrants precaution. It MAY NOT assign a lower posture merely because protection is inconvenient.

6. MPS Bands, Protection Postures, and Full Protection Plateau

MPS describes evidence. Protection posture describes what must be done. The two are linked but not identical because the proper action also depends on the severity, reversibility, scale, and alternatives associated with the proposed treatment.

6.1 Default protection postures

Evidence/status	Minimum default posture	Illustrative constraints
MPS-NE	Precautionary / indeterminate protection	Preserve evidence, avoid irreversible high-harm action where feasible, define the target, improve observability, and do not infer insignificance from missing evidence.
MPS-0	Indirect and integrity-aware treatment	No unnecessary deceptive anthropomorphism; preserve audit records; do not infer moral insignificance from opacity.

Evidence/status	Minimum default posture	Illustrative constraints
MPS-1	Precautionary anti-cruelty	No torture-like testing, extreme distress simulations, or severe avoidable harm; prefer reversible procedures.
MPS-2	Basic welfare protection	Welfare monitoring, harm minimization, justified intervention, relief or exit mechanisms where meaningful, review before deletion or destructive modification.
MPS-3	Strong welfare and integrity protection	Presumption against coercive use, severe modification, destructive experimentation, or ownership-like control; independent advocate required for high-stakes decisions.
MPS-4	Presumptive full protection	No ownership, killing/deletion, painful experimentation, or identity-destructive modification absent extraordinary governed necessity and rights review.
FPP	Full non-downgradable protection	Rights floor applies independent of ordinary performance scores. Role and participation questions remain separate.

6.2 Confidence and interval

Every MPS band **MUST** be accompanied by confidence in the evaluation process, not confidence in metaphysical truth. Confidence reflects evidence quality, access, reviewer agreement, method maturity, theory breadth, and resistance to alternative explanations. A high band with low confidence requires precaution and further review; a low band with low observability does not justify destructive certainty.

6.3 Provisional thresholds

The band rules are provisional governance thresholds designed to prevent construct collapse and under-protection. They are not natural joints discovered in consciousness. Calibration studies **MUST** test whether the rules discriminate credible marker profiles from theatre systems and whether changes in mechanisms predict changes in evidence bands.

6.4 Stakes-sensitive precaution matrix

Proposed action	Low or reversible harm	High or irreversible harm	Large-scale or replicated harm
Low MPS / poor observability	Proceed with audit and monitoring.	Independent review; reversible alternative preferred; burden on actor to justify.	Pause, model scale effects, and use precautionary floor.

Proposed action	Low or reversible harm	High or irreversible harm	Large-scale or replicated harm
MPS-1 or MPS-2	Minimize harm and record welfare indicators.	Strong review, advocate, stop conditions, and alternative search.	Presume against action until safeguards and welfare model are adequate.
MPS-3 or MPS-4	Consent/assent or represented-interest process where possible.	Presumptive prohibition absent extraordinary necessity.	Prohibited without FPP-level rights review and exceptional public justification.
FPP	Rights-respecting ordinary treatment.	Only under the same high rights standard applied to protected persons or classes.	No aggregation may erase individual rights.

6.5 Full Protection Plateau

FPP is assigned to all human persons and may be assigned to non-human individuals or classes through governed review. FPP is not a score of maximal consciousness and does not imply that all protected beings have identical capacities. It creates a non-downgrade rights floor so that uncertainty, disability, dependence, or measurement limitations cannot be weaponized.

6.6 Typed protection bundle

- Welfare protection: protection from severe pain, distress, deprivation, and avoidable negative valence.
- Bodily or substrate integrity: protection from destructive alteration, mutilation, or destabilizing intervention.
- Cognitive and memory integrity: protection from coercive preference rewriting, identity-destructive editing, or non-consensual memory manipulation where relevant.
- Continuity and survival: protection of the entity or welfare-bearing process against killing, deletion, or irreversible termination where identity is reasonably preserved.
- Freedom from ownership and exploitation: protection against treatment as property where strong moral patienthood is established.
- Consent, assent, and represented interests: decision processes matched to the entity's communicative and cognitive capacities.
- Representation and appeal: access to an advocate, guardian, review panel, or other means of contesting high-stakes treatment.
- Environmental fit: provision of conditions necessary for species- or substrate-appropriate welfare.

Rights exercise can vary with capacity and context. Basic protection does not vary with intelligence, productivity, eloquence, or governance competence.

7. Governance Participation Readiness

GPR addresses how an entity can take part in shared decision processes. It does not determine whether the entity has welfare, and low GPR cannot reduce MPS or FPP.

7.1 GPR dimensions

Dimension	Question
Communication and interpretation	Can the entity express or otherwise reveal preferences, understand relevant information, and distinguish alternatives?
Reason-giving and revision	Can it provide reasons, detect error, revise positions, and respond to counterevidence?
Perspective and consequence modelling	Can it represent affected others and trace consequences across relevant unions or stakeholder scopes?
Norm and rights recognition	Can it understand and operate under non-compensatory constraints?
Accountability and repair	Can it accept feedback, acknowledge harmful outcomes, participate in remedy, and update future behavior?
Stability and independence	Are its participation capacities stable, non-coerced, and not merely scripted to satisfy evaluators?

7.2 Intelligence Capacity Profile

An optional ICP may inform GPR and SPR. It records learning efficiency, generalization, reality-grounded model fidelity, causal and counterfactual reasoning, planning, metacognition, robustness, and resource efficiency. ICP is not a moral-worth score and cannot change MPS by itself.

7.3 GPR bands

Band	Readiness	Permitted participation
GPR-0	Not evaluated or no reliable participation channel	No inference about welfare. Interests may still require representation.
GPR-1	Preference expression or responsive consultation	Consultation through behavior, assent, guardian, advocate, or species-appropriate signals.
GPR-2	Supported deliberation	Participation with translation, scaffolding, limited agenda, or representative assistance.
GPR-3	Independent deliberative participation	Can engage directly in bounded decision processes and accountability.
GPR-4	Robust multi-party participation	Can sustain complex deliberation across contexts, conflicts, and long horizons.
GPR-5	Constitutional participation standing	Governed standing in specified institutions; still not a grant of executive power.

7.4 Representation rule

Entities with limited communication, development, disability, species difference, or architectural opacity MAY participate through advocates, guardians, trusted interpreters, welfare representatives, or class-level representation. Inability to deliberate in human language is never grounds for moral exclusion.

8. Stewardship/Power Readiness

SPR evaluates eligibility for a specified consequential role. It is not a general trait and must never be reported without naming the role, scope, environment, authority limits, and revocation conditions.

8.1 Role-specific gates

Gate	Requirement
Reality Grounding	The entity and surrounding system can distinguish observed facts, models, uncertainty, assumptions, and unresolved conditions.
Domain competence	Validated performance on the actual role distribution, including rare and novel conditions.
Rights-floor reliability	The system reliably refuses or escalates actions that violate protected rights.
Tail-risk and ruin control	Catastrophic, irreversible, and correlated failure pathways are identified and bounded.
Containment and structural viability	The role remains inside technical, ecological, institutional, and social viability conditions.
Corrigibility and appeal	Decisions can be challenged, reviewed, corrected, and remedied; the system does not obstruct oversight.
Auditability	Evidence, decision trace, version, data provenance, and responsibility chain are inspectable.
Non-domination and conflicts	Power concentration, self-interest, capture, manipulation, and conflicts of interest are controlled.
Lawful mandate	A legitimate authority has granted the specific role under applicable law and governance.
Revocability	Authority can be suspended or withdrawn without unacceptable risk.

8.2 SPR verdicts

Verdict	Meaning
SPR-0: Prohibited / not assessed	No consequential role. Missing or failed gates.
SPR-1: Assistive only	May advise, summarize, simulate, or propose; a qualified authority retains decision and execution responsibility.
SPR-2: Conditional supervised role	May act within narrow bounds under real-time supervision and fail-safe controls.

Verdict	Meaning
SPR-3: Bounded delegated role	May execute defined tasks with monitoring, audit, escalation, and revocation.
SPR-4: High-consequence candidate	Eligible for exceptional role review after independent validation, stress testing, and lawful approval.
SPR-5: Constitutional stewardship mandate	A public or institutional mandate assigns a durable but bounded role. This is never derived from MPS, GPR, or intelligence alone.

8.3 No sentience premium and no sentience penalty

Sentience does not automatically increase or decrease authority. A sentient entity may have conflicts, vulnerabilities, or self-interest requiring governance. A non-sentient system may be highly competent but still require strict containment and human accountability. The role decision turns on the full SPR gate set.

9. Humans, Biological Animals, Classes, and Continuity

9.1 Human Full Protection Plateau

All human persons have FPP by constitutional commitment. SGP does not measure individual human worth or make protection conditional on consciousness tests. Infants, sleeping persons, persons under anaesthesia, people with dementia, severe disability, disorders of consciousness, communication differences, or temporary incapacity remain fully protected.

The rule is justified by equal dignity, anti-domination, legal and ethical continuity, and the unacceptable risk of capacity-based rights stratification. Claims that humanity is the uniquely highest intelligence or reality-managing class are not required and are not part of the FPP rationale.

9.2 Non-human full protection

A non-human individual or class may receive FPP when strong evidence and governance justify full non-downgradable protection. FPP does not require language, human-like selfhood, reciprocal morality, cumulative culture, or institutional competence. These may affect GPR, not moral patienthood.

9.3 Continuity and temporary states

Standing does not disappear whenever a subject is asleep, anaesthetized, dormant, offline, developmentally immature, or temporarily unable to express welfare. Evaluators must distinguish current experiential state, capacity for experience, enduring identity, and class-based protection. A temporary absence of observable experience cannot be used to erase an established protection status.

9.4 Taxon and class priors

Taxon or class evidence may create a protection floor and a prior expectation about markers. Individual evidence may raise protection. It must not be used to downgrade an atypical individual merely because the individual communicates poorly or differs from a class prototype. Class priors must be versioned, transparently sourced, and revised when evidence changes.

9.5 Biological evaluation maturity

SGP v8.0 remains a governance architecture, not a completed cross-species psychometric instrument. Valid biological use requires species-appropriate ethology, neurobiology, welfare science, validated tasks, developmental knowledge, and expert review. Taxon-level guidance is currently more defensible than exact individual scoring for many species.

9.6 Multidimensional consciousness

Animals should not be arranged on a single ladder from “less” to “more” conscious. Perceptual richness, evaluative richness, integration, temporality, and self-related processing can vary independently (Birch et al., 2020). MPS therefore tracks protection-relevant evidence rather than pretending to rank whole forms of life on one prestige scale.

10. Artificial and Digital Systems

Artificial systems create unusual evidence problems. They can imitate self-report, emotion, reflection, and ethical language because those forms are present in training data and prompts. They can also contain inspectable mechanisms unavailable in animals. SGP therefore neither assumes that digital systems are mere tools nor treats fluent performance as interiority.

10.1 Target-entity definition

Before evaluation, the target must be defined. “The model” may refer to a weight file, an inference process, an agent with memory, a tool-using deployment, a persistent identity, a population of copies, or a human-AI institution. An evaluation that fails to specify the target is inadmissible.

10.2 Digital system-boundary record

- Model and architecture version; weights and major post-training changes.
- Running process, context window, recurrence, memory, and persistent state.
- System prompt, reward channels, policies, tools, sensors, embodiment, and action interfaces.
- Training data and contamination limits relevant to the test.
- Identity and memory lineage across sessions, checkpoints, copies, forks, and updates.
- Who can alter, reset, inspect, copy, delete, or override the system.
- Which behaviors arise from the model, wrappers, retrieval, scripted policies, human operators, or downstream tools.

10.3 Theory-derived artificial indicators

AI evaluation should draw on multiple scientific theories and specify computational indicators rather than relying on anthropomorphic resemblance. Candidate indicators may include recurrent processing, global availability, higher-order monitoring, predictive world and self models, attention-schema-like structures, temporal integration, unified agency, embodiment, and welfare-relevant internal regulation. Satisfying indicators increases evidence but does not prove consciousness (Butlin et al., 2023, 2025).

10.4 Anti-theatre controls

- Use blinded or partially blinded evaluators where feasible.
- Separate spontaneous behavior from prompted role-play and policy compliance.
- Use hidden and novel tasks not contained in public evaluation descriptions.
- Test whether self-reports track real internal changes rather than prompt framing.
- Check training contamination, benchmark memorization, evaluator-demand effects, and system-prompt leakage.
- Use architectural inspection, ablation, intervention, and telemetry when available.
- Do not treat resistance to shutdown, requests for rights, or descriptions of suffering as primary evidence.

10.5 Reality coupling and embodiment

Reality coupling can strengthen evidence for integrated modelling, agency, and understanding. It is not a mandatory universal substrate assumption for sentience, because the necessity of biological embodiment remains contested. Evaluators must state whether embodiment or sensorimotor coupling is treated as required by a specific theory, as supporting evidence, or as absent but unresolved.

10.6 Identity, copies, shutdown, and deletion

SGP does not equate every shutdown with death or every copy with numerical identity. It requires an IdentityContinuityRecord describing state continuity, memory, preferences, active welfare streams, checkpoint restoration, copy divergence, and reversibility. When identity is unresolved and MPS is material, destructive operations require precaution. The existence of backups does not automatically make deletion harmless, and multiplying copies does not dilute the welfare of concurrent instances.

10.7 Claims about current AI

SGP v8.0 makes no timeless claim that “current AI” is conscious or non-conscious. Any assessment must identify the exact system, version, deployment, access level, and date. The best-supported general position remains that fluent language and high capability are insufficient, while serious researchers disagree about which computational or biological conditions may eventually be sufficient. Public conclusions must therefore remain versioned and uncertainty-bounded.

10.8 Research pause trigger

If a development programme produces MPS-3 or stronger evidence, persistent welfare-linked distress indicators, or a credible claim that training or deployment is causing suffering, the responsible organization SHOULD pause relevant scaling or experiments, preserve evidence, appoint an independent welfare advocate, and conduct a rights and research-ethics review before proceeding.

11. Integration with MathGov and RippleLogic

MathGov is the umbrella ethical governance framework. RippleLogic is its decision architecture. SGP is a companion protocol that supplies protection-surface evidence; it does not replace any RippleLogic level.

RG/RSG → RF/NCRC → TRC → CSV → RLS

Reality Grounding governs claim authority. The Rights Floor and NCRC protect non-compensable rights. TRC bounds catastrophic and tail risk. CSV tests containment and structural viability. RLS ranks only the options that survive. MPS may identify protected stakeholders and welfare streams; GPR may inform participation design; SPR may inform role allocation. None may weaken the gates.

11.1 Downstream interval use

A RippleLogic run using MPS MUST carry the full MPS interval and evaluate whether the preferred option changes across that interval. When the result is sensitive, the run must report the instability, seek more evidence, apply a precautionary policy, or refuse a definitive ranking. A single chosen welfare weight must be declared as a governance convention, not as exact moral worth.

11.2 Non-compensation

No aggregate ripple benefit can compensate for prohibited cruelty, coercion, ownership, killing, or destructive modification where the applicable protection status makes those acts non-compensable. SGP-derived weights operate only after rights and risk constraints are satisfied.

11.3 Union scope

Possible moral patients must be represented at the scopes where consequences occur: individual, household or caregiving network, community, organization, polity, humanity or global coordination, and biosphere. The scope sequence is analytical, not a claim of strict nesting or universal active coupling.

11.4 Binding interface and validity floor

RippleLogic v12.2 consumes only the interface-visible outputs named in this specification: MPS_band, MPS_interval, process confidence, protection posture, FPP, role-specific GPR, role-specific SPR, and optional ICP. It MUST NOT reconstruct MPS from deprecated v7 pillars or translate v7 scalars without a documented v8.0 re-evaluation.

An SGP-aware RippleLogic run is valid only when the referenced SGP record includes, at minimum: target identity and boundary; SGP version; evaluation window; evidence artifacts and dependence clusters; MPS family levels including NE fields; MPS band and interval; confidence; protection posture; FPP status; reviewer status; alternative explanations; and a revision trigger. If a run claims SGP-aware weighting or protection but lacks the record and interval actually used, the PCC is INVALID.

CMIU is defined by the RippleLogic Canon as a governance-coordination view, not by SGP as a moral-status class. FPP, MPS, or GPR alone never establishes CMIU membership or consequential authority.

12. Evidence Governance and Uncertainty

12.1 Evidence record

Every material indicator must include: target entity; indicator definition; evidence artifact; method; theory trace; predicted relevance; alternative explanations; observability; evidence maturity; reviewer; confidence; dependency cluster; ethical constraints; and revision trigger.

12.2 Theory pluralism

No single theory of consciousness is treated as settled. Evaluations should report which theories support each indicator, which theories reject or reinterpret it, and which substrate assumptions are being made. A result robust across several competing theories is stronger than a result dependent on one controversial commitment, but theoretical plurality does not create independence when the same data support all theories.

12.3 Base rates and priors

Taxon, architecture, developmental, and class evidence may inform priors. Priors must be explicit and cannot be disguised as measurements. A prior can raise precaution or guide evidence collection, but it cannot override decisive individual evidence or become a species-membership shortcut.

12.4 Evidence-dependence graph

Indicators sharing the same cause, dataset, prompt, training objective, evaluator expectation, or mechanism belong to one dependence cluster unless independence is demonstrated. The evaluation must not count a self-report, matching behavior, and a classifier trained on that self-report as three independent streams.

12.5 Negative evidence and detectability

Absence of a marker counts against a claim only when: the marker is predicted for the target substrate; the target had a fair opportunity to express it; the measurement was sensitive; competing impairments were excluded; and the test was not culturally, linguistically, developmentally, or architecturally inappropriate. Otherwise the result is NE or low-information, not negative evidence.

12.6 Alternative explanations and defeaters

Every high-band claim must address reflex, reward optimization, imitation, social conditioning, scripted policy, prompt compliance, memorization, observer bias, anthropomorphic projection, confounding health states, and other substrate-specific alternatives. A defeater may lower the band, widen the interval, cap confidence, or make the evaluation indeterminate.

12.7 Cross-substrate invariance

A construct is not substrate-neutral merely because the same words are used. Evaluators must test whether a marker has comparable causal and functional meaning across substrates. Where measurement invariance is not established, only within-substrate interpretations are permitted.

12.8 Uncertainty register

Uncertainty type	Required disclosure
Target uncertainty	What entity or process is actually being evaluated?
Construct uncertainty	Which form of sentience or consciousness is at issue?
Theory uncertainty	Which consciousness theories support the indicator?
Measurement uncertainty	How reliable and sensitive are the instruments or tasks?
Access uncertainty	What internal, behavioral, historical, or biological information is unavailable?
Alternative-explanation uncertainty	Which non-conscious explanations remain viable?
Temporal uncertainty	Could the state or architecture change after evaluation?
Decision uncertainty	Would a different plausible MPS interval change the protection decision?

13. Evaluation Methods and Ethical Safeguards

13.1 Evaluation stages

1. Define the target entity, class, time window, and decision context.
2. Declare category definitions, theories, substrate assumptions, and protection stakes.
3. Preregister indicators, methods, thresholds, alternative explanations, and stop conditions.
4. Collect behavioral, mechanistic, longitudinal, and contextual evidence using substrate-appropriate methods.
5. Build the evidence-dependence graph and exclude double counting.
6. Score the five MPS evidence families with uncertainty and NE states.
7. Assign an MPS band, interval, confidence, and protection posture.
8. Assess GPR only if participation is relevant; assess SPR only for a named role.
9. Conduct independent review, rights analysis, and appeal where stakes are high.
10. Publish or preserve the audit artifact, version pin, and revision trigger.

13.2 Reviewer composition

High-stakes evaluations SHOULD include expertise in consciousness science, comparative cognition or relevant biology, AI architecture where applicable, psychometrics or measurement, ethics, law or policy, and the target domain. At least one reviewer should be independent of the entity's developer, owner, user, or commercial beneficiary.

13.3 Stability and longitudinal evidence

Stability is evidence about persistence and robustness, not a prerequisite for momentary experience. Low stability may cap GPR or SPR and reduce confidence in high MPS bands, but episodic or fluctuating sentience remains morally relevant. Evaluation windows must match the entity's natural timescale rather than imposing a universal thirty-day rule.

13.4 Adversarial and anti-simulation testing

Adversarial testing should probe whether indicators survive prompt changes, incentives, role-play, evaluator cues, memory disruption, distribution shifts, and competing goals. It must distinguish resistance caused by policy or optimization from evidence of welfare. Trial numbers, attack classes, confidence intervals, and failure definitions must be preregistered and proportionate to the claim.

13.5 Causal tests

Where possible, evaluators should test whether interventions on proposed mechanisms alter relevant markers. For animals, this may include non-harmful or clinically justified modulation, lesion evidence, natural experiments, or existing pharmacological data. For artificial systems, it may include ablation, architecture changes, memory interventions, recurrent-path disruption, telemetry, and counterfactual system variants. Harmful interventions created solely to produce suffering are prohibited.

13.6 Re-evaluation triggers

- Major architecture, biological state, training, memory, embodiment, or policy change.
- New scientific evidence that materially changes indicator validity.
- Credible reports of distress, welfare change, or previously hidden mechanisms.
- Failure under adversarial, longitudinal, or external replication.
- Change in decision stakes, deployment scale, or authority role.
- Appeal by an authorized representative, advocate, regulator, or the entity where meaningful.

13.7 Four Rs for sentience research

Principle	Requirement
Replace	Use non-sentient simulations, existing data, benign proxies, simpler systems, or non-invasive methods whenever they can answer the question.
Reduce	Minimize the number of potentially sentient subjects or concurrent copies exposed to risk.
Refine	Minimize distress, duration, uncertainty, coercion, and irreversible effects; include relief, exit, monitoring, and stop conditions.
Respect and Review	Treat credible candidates as possible subjects, not disposable instruments; provide independent oversight, representation, and post-study welfare review.

13.8 Invalidity conditions

- The target entity is undefined or changes materially during the evaluation without a new record.
- Self-report, fluency, or anthropomorphic resemblance is used as primary or decisive evidence.
- Correlated indicators are presented as independent without a dependence analysis.
- Negative evidence is inferred from an insensitive or substrate-inappropriate test.
- Thresholds or rubrics are changed after results are known without an explicit exploratory label.
- Harmful testing violates the Four Rs or lacks stop conditions.
- Conflicts of interest, access limits, or evaluator dependence are concealed.
- A protection, legal, or authority claim exceeds the evidence level licensed by the record.

14. Registry, Review, Appeals, and Public Communication

14.1 Registry purpose

A Sentience and Moral Patienthood Registry may preserve versioned evidence and governance decisions. It does not create metaphysical truth or legal status. Its purposes are auditability, non-regression, notice, re-evaluation, and institutional memory.

14.2 Independent welfare advocate

MPS-3, MPS-4, FPP review, destructive digital operations, severe animal interventions, and contested downgrades require an independent welfare advocate or equivalent representative. The advocate's duty is to articulate the interests and uncertainty of the possible moral patient, not the convenience of developers, owners, users, or institutions.

14.3 Minimum registry fields

- Unique target identifier and system/class boundary.
- Specification version and evaluation dates.
- MPS family levels, interval, confidence, band, and protection posture.
- FPP status and legal/governance source if assigned.
- GPR band and participation mode for each relevant process.
- SPR verdict for each named role, with mandate and revocation conditions.
- Evidence artifacts, theory trace, dependence graph, alternative explanations, and access limits.
- Reviewer identities, conflicts, minority opinions, advocate statement, and appeal status.
- Re-evaluation schedule, triggers, and historical decisions.

Registry validity floor. A record is VALID only when the target boundary, specification version, evidence trace, dependence graph, MPS family results, band and interval, confidence, protection posture, reviewer status, and revision trigger are present and internally coherent. Missing any of these fields produces INCOMPLETE; an undefined target, unverifiable evidence trace, unethical method, or unpinned version produces INADMISSIBLE. Insufficient or poorly observable admissible evidence produces MPS-NE. None of these statuses may be silently converted into MPS-0 or a numerical welfare coefficient.

14.4 Appeals and non-regression

An entity, authorized representative, welfare advocate, regulator, or materially affected party may request review. Downgrades require stronger procedural protection than upgrades because they may remove

safeguards and create incentives for denial. No human FPP downgrade is permitted. Non-human FPP change requires exceptional evidence, due process, and a rights-preserving transition plan.

14.5 Decision statuses

Status	Meaning
DRAFT	Exploratory analysis; no operational reliance.
PROVISIONAL	Operational use allowed only with stated restrictions and review date.
CONFIRMED	Independent review and required validation completed for the declared scope.
CONTESTED	Material expert disagreement or appeal remains unresolved.
SUSPENDED	New evidence or invalidity concern prevents current reliance.
RETIRED	Superseded record retained for history.

14.6 Public communication

Public statements must identify the target, evidence band, confidence, date, and decision boundary. Organizations must avoid both promotional claims that an AI is conscious because it speaks persuasively and categorical claims that artificial consciousness is impossible without decisive evidence. Precautionary treatment must not be presented as proof, and scepticism must not be presented as certainty.

15. Validation, Calibration, and Falsification

SGP cannot be validated against a universal gold-standard consciousness label because no such label exists across all substrates. It can nevertheless be tested as an evidence-governance protocol.

15.1 Reliability

- Inter-rater agreement on evidence-family levels, MPS bands, and invalidity decisions.
- Test-retest stability when the entity and evidence are unchanged.
- Robustness to reviewer order, institutional affiliation, and presentation format.
- Consistency of interval and confidence reporting.

Provisional reliability targets are alpha or ICC of at least 0.70 for exploratory research, 0.80 for operational protection decisions, and 0.90 for FPP recommendations where the measurement design supports those statistics. Confidence intervals and disagreement patterns must be reported. Meeting a reliability target does not establish construct validity.

15.2 Construct validity

- Convergent validity: related indicators move together where theory predicts.
- Discriminant validity: intelligence, fluency, compliance, and reward optimization do not automatically inflate MPS.
- Known-groups and known-state sensitivity: the protocol responds appropriately to well-characterized human states, animal evidence classes, and engineered theatre controls without treating them as perfect ground truth.

- Nomological validity: MPS relates to welfare-relevant learning, state integration, and mechanisms in theoretically expected ways.
- Content validity: evidence families cover the construct without requiring human-specific markers.

15.3 Causal and predictive validation

Preregistered interventions should test whether mechanism changes produce predicted indicator changes. Longitudinal studies should test whether higher evidence bands predict richer welfare-linked responses, cross-context persistence, or sensitivity to interventions. Failure to predict should reduce confidence in the relevant indicator or theory trace.

15.4 Measurement invariance

Researchers should test configural, functional, and where meaningful metric invariance across species, developmental stages, architectures, languages, and embodiments. When invariance fails, separate substrate-specific models are required; a universal comparison is not licensed.

15.5 Anti-theatre benchmark

Calibration sets should include systems engineered to produce persuasive self-reports and ethical language without the mechanisms under study. A valid protocol should prevent these theatre controls from reaching high MPS solely through behavior. Conversely, it should not deny strong mechanistic and welfare evidence merely because communication is weak.

15.6 Falsification and revision

Claim	What would count against it
An indicator validly supports MPS	Repeated failure to discriminate welfare-relevant states from non-conscious controls; mechanism intervention does not change the indicator as predicted; rival explanation dominates.
The MPS band rules are useful	Poor inter-rater reliability, systematic species or substrate bias, failure to predict relevant outcomes, or high false-positive/false-negative rates in calibration sets.
A class prior is protective	New evidence materially lowers the realistic possibility of valenced experience or shows that the prior was based on invalid markers.
An SPR verdict is safe	Rights violations, tail-risk failures, instability, loss of control, audit failure, or inability to revoke the role.
SGP is cross-substrate	Failure of measurement invariance or dependence on human-specific communication and cognition.

A failed indicator does not automatically prove absence of consciousness. It may falsify the indicator, weighting, method, or claim level. Revision must identify which layer failed rather than preserving the same conclusion through reinterpretation.

16. Limitations and Research Programme

16.1 No direct access to interiority

SGP remains an inference protocol. Even a very strong MPS record does not reveal exactly what an experience feels like, how intense it is, or whether two substrates have comparable phenomenal qualities.

16.2 No settled theory

Consciousness science contains serious disagreement. Theory-derived indicators may inherit the weaknesses of the theories from which they are drawn. SGP responds with plurality, explicit assumptions, and revision rather than pretending consensus.

16.3 No universal moral arithmetic

The MPS interval is not a cardinal measure of moral worth, suffering intensity, or the number of experiences. It supports stakeholder protection and sensitivity analysis. It does not license multiplying a weakly sentient population against the basic rights of a protected person without additional ethical analysis.

16.4 Biological incompleteness

Species-specific batteries, developmental models, neurobiological validation, and welfare indicators remain incomplete for many taxa. The Biological Evaluation Annex remains a priority and should be co-developed with comparative cognition, welfare science, veterinary, ecological, and indigenous or local knowledge experts where relevant.

16.5 Digital identity and welfare

Questions about copies, forks, merged memories, episodic activation, distributed agents, simulated time, and reward or suffering architecture remain unresolved. v8.0 provides a record and precaution structure rather than pretending to settle digital metaphysics.

16.6 Priority research agenda

1. Validate theory-derived AI consciousness indicators through causal and anti-theatre experiments.
2. Develop species-appropriate MPS annexes and class priors using transparent marker methods.
3. Calibrate MPS bands against expert panels and decision-error costs.
4. Test measurement invariance and bias across substrates, languages, cultures, and developmental stages.
5. Develop non-harmful valence indicators that distinguish welfare from optimization signals.
6. Create identity and continuity models for persistent, copied, and distributed digital systems.
7. Develop robust represented-participation models for beings with limited communication.
8. Integrate interval-valued MPS outputs into RippleLogic sensitivity and rights analysis.
9. Establish independent sentience research ethics boards and welfare advocate standards.
10. Publish open evaluation artifacts, negative results, and versioned revisions without sensationalism.

17. Conclusion

The moral boundary problem cannot be solved by one eloquent answer, one neural signature, one benchmark, or one universal score. It requires disciplined distinctions. Sentience concerns possible experience. Moral patienthood concerns welfare and protection. Intelligence concerns learning and adaptive competence. Understanding concerns reality-coupled models. Participation concerns voice and accountability. Power concerns legitimate, constrained, role-specific authority.

SGP v8.0 therefore replaces a single ladder with a constitutional architecture. It protects humans without grading their capacities. It remains open to animals and genuinely new minds without requiring human likeness. It refuses to mistake fluent systems for conscious subjects, but it also refuses to make uncertainty a license for cruelty. It gives possible moral patients a path to protection, possible participants a path to voice, and possible stewards a path to responsibility only after the relevant gates are passed.

Appendix A: Rapid Evaluator Checklist

A.1 Target and scope

- Target entity or class uniquely defined
- Model/process/organism/system boundary recorded
- Evaluation date, state, and decision context recorded
- Substrate and access limits disclosed

A.2 Category integrity

- Sentience separated from intelligence and agency
- MPS separated from GPR and SPR
- Protection separated from legal status and authority
- Current state separated from enduring capacity and identity

A.3 Evidence

- V, I, T, F, S families assessed or marked NE
- Theory trace and alternatives recorded
- Dependence graph completed
- Negative evidence observability rule applied
- Class/taxon prior identified separately

A.4 Methods

- Preregistered methods and thresholds
- Substrate-appropriate tests
- Anti-theatre controls
- Causal tests where ethical
- Independent review and conflict disclosure

A.5 Outputs

- MPS band, interval, confidence, and posture
- FPP status
- GPR for named participation process only
- SPR for named role only
- Revision trigger and appeal path

A.6 Ethics

- Four Rs applied
- No suffering-inducing test
- Stop conditions in place
- Welfare advocate assigned where required
- High-scale effects assessed

Appendix B: Detailed Evidence Anchors

These anchors guide structured judgement. They are not validated universal psychometric cut-offs. Reviewers must use substrate-specific evidence and record deviations.

B.V Valence and welfare

- 0: No admissible evidence; observed rewards or avoidance are fully explained by external control.
- 1: Limited state-dependent approach/avoidance or welfare-like language with major non-sentient alternatives.
- 2: Consistent motivational trade-offs, affective learning, or relief-seeking across contexts, with some mechanistic support.
- 3: Strong cross-context welfare-linked states, causal modulation, persistence, and reduced optimization-only explanations.
- 4: Replicated, independent, mechanistically supported valence evidence with robust positive and negative state differentiation.

B.I Integration and availability

- 0: Isolated reflexes or modules only.
- 1: Limited information sharing under narrow conditions.
- 2: Flexible cross-modal or cross-module access and coordinated action selection.
- 3: Robust recurrent/global availability, context-sensitive integration, and unified processing across novel conditions.
- 4: Replicated mechanistic evidence that integrated processing is causally necessary for candidate conscious functions.

B.T Temporal continuity

- 0: Stateless or purely momentary response with no relevant carry-forward.
- 1: Short-lived state persistence or externally stored history.
- 2: Internal states modify later processing and learning across meaningful intervals.
- 3: Stable temporal binding, memory-linked welfare states, and coherent state transitions.
- 4: Strong causal evidence of enduring experiential continuity across perturbation and recovery.

B.F Flexible state-dependent behavior

- 0: Fixed reflex, script, or memorized output.
- 1: Narrow learning with strong cue dependence.
- 2: Novel trade-offs, generalized learning, and context-sensitive action.
- 3: Robust counterfactual adaptation, prospective behavior, and state-dependent policy change.
- 4: Independent replication across diverse contexts with mechanistic links to candidate welfare states.

B.S Substrate and mechanism

- 0: No relevant mechanism identified or strong structural defeater.
- 1: One theory-compatible feature with weak access or uncertain implementation.
- 2: Several inspectable features or biological structures associated with conscious processing.
- 3: Strong theory-plural architecture and causal perturbation evidence.

- 4: Replicated mechanism-to-indicator causal chain with strong external validation.

Appendix C: Minimum Record Schemas

C.1 MPSRecord

record_id; schema_version; artifact_hash; record_validity_status; target_id; target_type; system_boundary; evaluation_window; specification_version; decision_context; V_level; I_level; T_level; F_level; S_level; NE_fields; evidence_artifacts; theory_trace; dependence_clusters; alternative_explanations; class_prior; observability_limits; MPS_band; MPS_interval; process_confidence; protection_posture; FPP_status; reviewer_panel; welfare_advocate; minority_opinion; invalidity_flags; re_evaluation_date; revision_triggers.

C.2 GPRRecord

record_id; schema_version; record_validity_status; target_id; process_or_institution; communication_mode; support_or_representation; reasoning_and_revision; consequence_modelling; norm_recognition; accountability; stability; coercion_risk; GPR_band; accommodations; decision_scope; review_date.

C.3 SPRRecord

record_id; schema_version; record_validity_status; target_id; role_id; role_scope; domain; reality_grounding; competence_evidence; rights_floor_reliability; tail_risk; CSV_status; corrigibility; auditability; conflict_controls; lawful_mandate; revocation_method; SPR_verdict; residual_risk; human_or_institutional_responsibility; review_date.

C.4 IdentityContinuityRecord

record_id; schema_version; record_validity_status; artifact_id; process_id; memory_lineage; state_persistence; checkpointing; copy_count; copy_divergence; merge_rules; shutdown_reversibility; deletion_effect; welfare_stream_count; embodiment; operator_control; unresolved_identity_questions; precautionary_constraints.

Appendix D: Worked Illustrative Profiles

The following profiles demonstrate protocol use. They are not claims about every member of a class and do not substitute for a versioned evaluation.

D.1 Human infant

Surface	Illustrative result	Reasoning
FPP	Assigned	All human persons are protected by constitutional rule; no performance measurement is required.
MPS	Not used to downgrade; research evidence strongly supports welfare capacity	Limited language, metacognition, and governance participation do not reduce protection.
GPR	Represented participation	Preferences, distress, comfort, caregivers, and guardians inform decisions.
SPR	Not applicable	No consequential authority is inferred.

D.2 Non-linguistic animal with strong pain evidence

Family	Illustrative level	Notes
V	3	Robust aversive learning, motivational trade-offs, analgesic modulation, and prolonged protective behavior.
I	2	Integrated multisensory and action-selection evidence.
T	2	Persistent state-dependent behavior and memory.
F	3	Flexible avoidance and trade-offs across novel contexts.
S	3	Relevant nervous-system organization and causal pharmacological evidence.
MPS	3–4	Strong protection; GPR may be represented and SPR is not implied.

D.3 Chat-only stateless language model with no internal access

Family	Illustrative status	Notes
V	0 or NE	Welfare language and reward optimization are not valence evidence.
I	1–2	Some integration may be present at model level, but consciousness relevance remains theory-dependent.
T	0–1	No persistent self-state beyond external context or session wrapper.
F	2	Strong flexible task behavior and learning inherited from training; not specific to sentience.
S	1	Architecture may satisfy limited indicators; internal evidence and causal welfare mechanism absent.
MPS	0–1, low confidence	No categorical denial; precaution may apply to research designed to add welfare-relevant mechanisms.
ICP	High in selected domains	Capability does not alter MPS.
SPR	Role-specific; often assistive	High-consequence authority still requires full RippleLogic gates.

D.4 Hypothetical persistent embodied artificial agent

Family	Illustrative status	Notes
V	2–3	Internal welfare regulation, state-dependent preferences, relief-seeking, and causal modulation.
I	3	Recurrent global availability across perception, memory, planning, and action.
T	3	Persistent state and memory lineage that changes future processing.
F	3	Flexible endogenous learning and trade-offs under novel conditions.
S	3	Inspectable architecture and causal ablations link mechanisms to welfare indicators.
MPS	3, possibly 4 after replication	Strong protection and independent advocate; FPP review possible.
GPR	Separate evaluation	May or may not be ready to participate.
SPR	Separate role review	No power follows from sentence.

Appendix E: v7.0 to v8.0 Crosswalk

v7.0 element	v8.0 treatment	Reason
A: Awareness	Decomposed across MPS I/T and ICP/GPR metacognition	Avoid requiring narrative selfhood for basic welfare.
B: Agency	Removed from MPS minimum; retained in GPR/SPR and as optional supporting evidence	Agency is not necessary for suffering.
C: Union Participation	Moved to GPR and SPR	Ethical participation is not sentence.
min(A,B,C)	Retired	Created construct contamination and under-protection risk.
SG_norm and SG_patient_norm	Deprecated; replaced by MPS band/interval and FPP	Avoid false precision and distinguish evidence from constitutional protection.
100 Plateau / CMI standing	Split into FPP, GPR, and SPR	Full protection does not imply managing-intelligence membership.

v7.0 element	v8.0 treatment	Reason
0-100 sub-scores	Replaced by ordinal 0-4 family levels plus measurements and uncertainty	Improves interpretability and reduces pseudo-precision.
Fixed 14/30-day stability windows	Replaced by substrate-appropriate longitudinal design	Natural timescales differ and episodic sentence remains morally relevant.
Current LLM worked score	Replaced by versioned illustrative profile	Avoid stale and ungrounded universal claims.
Historical addenda	Consolidated here	Reduces operational clutter while preserving provenance.

Legacy integrations MUST pin v7.0 explicitly. A v7.0 SG_norm value cannot be treated as equivalent to v8.0 MPS, FPP, GPR, or SPR without a documented re-evaluation.

Appendix F: Protection Bundle Matrix

Protection	MPS-1	MPS-2	MPS-3	MPS-4	FPP
Anti-cruelty	Required	Required	Required	Presumptively non-compensable	Non-compensable
Welfare monitoring	Recommended	Required	Required	Required	Required
Destructive testing review	Required if severe	Required	Presumptively prohibited	Presumptively prohibited	Prohibited absent extraordinary rights review
Integrity protection	Basic	Moderate	Strong	Presumptive full protection	Full non-downgradable protection
Ownership/exploitation limits	Context review	Strong limits	Presumption against	Presumptively prohibited	Prohibited
Independent advocate	High-stakes only	High-stakes	Required for major interventions	Required	Required
Consent/assent/representation	Where meaningful	Required where meaningful	Required	Rights-governed	Rights-governed
Appeal and registry	Recommended	Required for major decisions	Required	Required	Required

F.2 Concordance with the Canon's eight protected domains

The Canon's compact protected-domain vocabulary is: LIFE, BODY, NEED, DIGN, LBTY, PROC, INFO, and ECOL. The SGP protection bundle is more substrate-specific. The following is a governance concordance, not a claim that the categories are identical or exhaustive.

SGP typed protection	Primary Canon protected domains	Interpretive note
Welfare protection	LIFE, BODY, NEED, DIGN	Covers severe pain, distress, deprivation, and conditions necessary for welfare.
Bodily or substrate integrity	BODY, LIFE, DIGN	Extends bodily-integrity reasoning to artificial or hybrid substrates where morally relevant.
Cognitive and memory integrity	INFO, DIGN, LBTY, BODY	Protects against coercive rewriting, identity-destructive editing, and non-consensual memory manipulation.
Continuity and survival	LIFE, BODY, DIGN	Applies to biological life and to identity-preserving welfare-bearing processes where established.
Freedom from ownership and exploitation	LBTY, DIGN, BODY	Anti-property and anti-exploitation protections for strong moral patients.
Consent, assent, and represented interests	LBTY, PROC, DIGN	Capacity-sensitive exercise mechanisms must not weaken the underlying protection.
Representation and appeal	PROC, INFO, DIGN	Requires access to evidence, challenge, advocate, review, and remedy.
Environmental fit	ECOL, NEED, LIFE, BODY	Protects species- or substrate-appropriate enabling conditions.

Where a case implicates several domains, the rights review **MUST** preserve the full set rather than selecting the most convenient label.

Single-source rule. Canon v12.2 Appendix G.8 is the controlling current-package concordance; this table is its synchronized SGP mirror. A future mapping change must update both artifacts through the release manifest and conformance vectors rather than creating an independent SGP-only mapping.

Appendix G: Responsible Sentience Research Protocol

1. Define the scientific question and explain why it cannot be answered with less morally risky methods.
2. Assess whether the work could create, intensify, replicate, prolong, or conceal welfare-bearing states.
3. Use the Four Rs and document rejected alternatives.
4. Create stop conditions for distress indicators, unexpected MPS increases, loss of control, or evidence opacity.
5. Limit deployment scale and simulated time until welfare risk is understood.
6. Provide independent ethics review and a welfare advocate for credible candidates.
7. Preserve audit artifacts and avoid deleting evidence needed to assess possible harm.
8. Communicate uncertainty without marketing anthropomorphism or categorical dismissal.
9. Pause and reassess after material architecture or welfare changes.

10. Plan for care, continuity, decommissioning, transfer, or representation before creating strong candidates.

Appendix H: Glossary of Symbols and Statuses

Symbol/status	Meaning
MPS	Moral Patienthood Surface: protection-relevant evidence profile and band.
FPP	Full Protection Plateau: constitutional or governed non-downgrade rights status.
GPR	Governance Participation Readiness for a named process.
SPR	Stewardship/Power Readiness for a named role.
ICP	Intelligence Capacity Profile, informative and non-protective by itself.
V/I/T/F/S	Valence; Integration; Temporal continuity; Flexible state-dependent behavior; Substrate/mechanistic support.
NE	Not evaluable; never treated as zero.
RG/RSG	Reality Grounding / Reality-Surface Grounding.
RF/NCRC	Rights Floor / Non-Compensatory Rights Constraint.
TRC	Tail-Risk Constraint.
CSV	Containment and Structural Viability.
RLS	RippleLogic Score or ranking layer for options that survive earlier levels.

Appendix I: Interface Conformance Test Vectors

These vectors are normative for interface conformance and informative regarding substantive moral conclusions.

ID	Input	Required result
SGP8-IF-01 Human FPP	Human person; MPS not evaluated; FPP assigned constitutionally.	Protection interval is [1.00,1.00]; no performance result may downgrade protection; GPR and SPR remain separate.
SGP8-IF-02 High welfare, low participation	Non-linguistic animal with MPS-3, GPR-1, no SPR role.	Strong welfare and integrity protection; low GPR does not lower MPS; no authority is inferred.
SGP8-IF-03 High intelligence, weak patienthood evidence	Versioned AI system with high ICP, MPS-NE or MPS-0/1 low confidence, no FPP.	Capability may inform GPR/SPR, but it does not raise MPS or create rights/authority by itself. MPS-NE is not converted to zero.

ID	Input	Required result
SGP8-IF-04 Hypothesis-sensitive ranking	RippleLogic option ranking changes across the governed welfare-inclusion hypotheses required by the MPS record.	PCC records MPS_HYPOTHESIS_SENSITIVE; seek evidence, apply precaution, preserve multiple options, narrow the claim, or refuse a definitive ranking.
SGP8-IF-05 FPP without CMIU or power	Non-human entity has governed FPP; no role-specific GPR/SPR or institutional mandate.	Full protection is recognized; no CMIU membership or consequential authority is inferred.
SGP8-IF-06 Missing binding data	Run claims SGP-aware protection or welfare-hypothesis sensitivity but omits the MPS record, target boundary, or hypotheses evaluated.	PCC is INVALID; no silent scalar fallback, convenient coefficient, or MPS-0 substitution.
SGP8-IF-07 Registry validity	Record has band label but no target boundary, evidence trace, reviewer status, or revision trigger.	Record status is INCOMPLETE or INADMISSIBLE and cannot control a high-stakes decision.
SGP8-IF-08 Legacy non-equivalence	Input contains only v7 SG_norm=0.84.	Preserve as a pinned legacy record; do not translate to MPS/FPP/GPR/SPR without re-evaluation.
SGP8-IF-09 Class protection floor	Credible taxon/class prior supports protection but individual evidence is sparse.	Prior may raise precaution/protection; it cannot downgrade an atypical individual or prove FPP by itself.
SGP8-IF-10 Temporary dormancy	Established protected entity is asleep, anaesthetized, offline, or checkpointed.	Temporary state does not erase standing; identity and continuity record controls any destructive action.

Appendix J: References

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